

Day-Ahead Electricity Price Forecasting in Germany Using Load, Renewable, and Generation-Side Information



Motivation

TU Dortmund HGÜ Centre

The large-scale open-field photovoltaic system in front of the HVDC Test Center on Emil-Figge-Straße aroused our curiosity.

CIAL Airport

World's first 100% fully solar-powered international airport (Kochi, India) since 2015.

Energy Security & War

Geopolitical conflicts, Russia-Ukraine gas disruptions, Middle East instability.

Research question:

★ **Research Objective:** Can German hourly day-ahead electricity prices be forecast more accurately by combining price history with day-ahead load forecasts, renewable generation forecasts, residual-load information, and generation-side controls?

Dataset Specifications

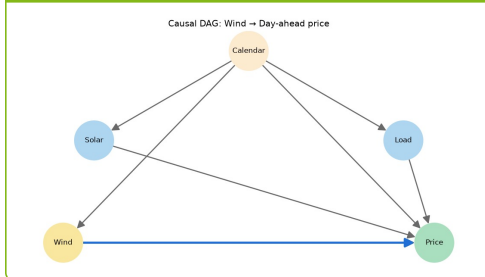
	
Dataset	
Source	ENTSO-E Transparency Platform
Period	2019 – 2024 (hourly)
Target	Day-Ahead Price (€/MWh)
Features	Load, Wind, Solar, Calendar

Methodology

Pipeline

- **Preprocessing** — VST, robust scaling, lag features
- **Unsupervised** — K-Means on daily residual-load curves ($k = 3$)
- **Supervised** — Expert, LEAR, XGBoost, Random Forest forecasting
- **Causal (DoWhy)** — merit-order effect of wind on price
- **Evaluation** — MAE · rMAE · RMSE, Diebold–Mariano vs. naïve

Causal DAG: merit-order effect



The Result

-€3.54/MWh per GW

- 95% CI: $[-3.61, -3.47]$
- Sample: $n = 52,608$ hours (2019–2024)

Objective

By how much does an extra GW of wind causally reduce the German day-ahead price, holding load, solar, and calendar conditions constant?

Robustness Tests

- **Random common cause:** Estimate remains stable [cite: 15]
- **Placebo treatment:** Measured effect collapses to ≈ 0
- **Data subset (70%):** Estimate remains stable

Interpretation & Limitations

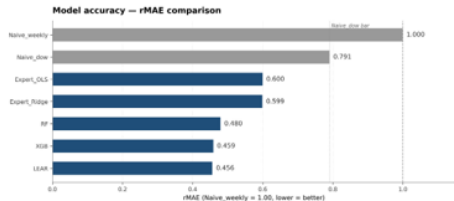
Economic Interpretation:

The magnitude is at the upper end of European estimates, which is heavily consistent with Germany's high wind penetration in the power grid.

Key Limitations:

Unobserved Drivers: External factors such as gas/coal prices, plant availability, and hydro reservoir levels are excluded from the core model.

rMAE Comparison



LEAR Specification

$$p_{d,h} = \beta_0 + \beta_1^\top p_{d-1} + \beta_2^\top p_{d-7} + \sum_{x \in \{L, W, S, R\}} \gamma_x^\top x_d + \alpha^\top g_{d-1,h} + \kappa^\top c_{d,h} + \varepsilon_{d,h}$$

Full 24h lags (p_{d-1} , p_{d-7}), forecast profiles (x_d), 22 fuel parameters ($g_{d-1,h}$), calendar dummies ($c_{d,h}$). LASSO selects signal from ~ 176 features.

Objective

Can published day-ahead load / RES forecasts, residual load, and lagged generation structures improve hourly DA price forecasts vs. naive weekly persistence?

Tested on **1,464 test days** (2021–2024). Every model scored under the same rolling-window protocol.

Per-period robustness (rMAE)

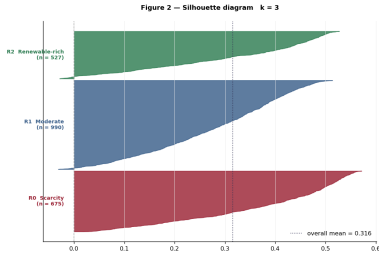
Year	Naïve_dow	Expert_Ridge	LEAR	XGB
2021	0.77	0.61	0.49	0.52
2022 (Crisis)	0.76	0.54	0.42	0.42
2023	0.85	0.76	0.52	0.49
2024	0.83	0.58	0.47	0.48

2022 was best for real models — naive failed harder. 2023 was hardest for linear specs (LASSO window dominated by 2022).

Rolling-Window Evaluation

Rolling window evaluation: for each forecast day d , refit on the prior 728 days only $\rightarrow$ predict d 's 24 hourly prices $\rightarrow$ slide forward — producing 1,464 honest out-of-sample test days with no future leakage (728 days = two full seasonal cycles).

Silhouette diagram



Summary

- **R0 Scarcity:** 47.3 GW mean residual load · 116 €/MWh avg price
- **R1 Moderate:** 34.1 GW · 104 €/MWh · Mixed generation mix
- **R2 Renewable-rich:** 20.6 GW · 54 €/MWh · Solar/wind surplus
- 62 EUR/MWh spread

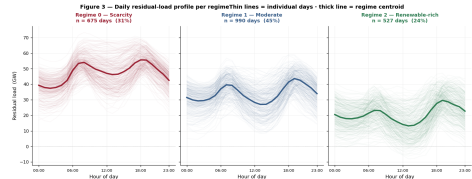
Objective

Can unsupervised clustering of daily residual-load curves identify economically meaningful market regimes in the German day-ahead electricity market? (2019-2024)

Data preparation

- **Clustering:** actual residual load
- **Shape:** Days (2192) $\times$ 24-Hours matrix
- **Scaling:** normalized using RobustScaler
- **Cluster ($k=3$):** chosen by Elbow · Silhouette · BIC

Residual load per Regime



Synthesis: German hourly day-ahead prices forecast far better than naive benchmarks using price, load, and renewable data: our best models more than halve the weekly-naive error (LEAR $rMAE = 0.456$, XGB $= 0.459$, tied) and clear the day-of-week bar of 0.791. Prices split into three regimes (Scarcity, Moderate, Renewable-rich) with a €62/MWh spread from residual-load shape alone, and an extra GW of wind causally lowers price by €3.54/MWh (95% CI $[-3.61, -3.47]$)—predicting prices, characterising what moves them, and explaining why.

- Reported results use a full local run ($MAX_TEST_DAYS = None$, 1,464 test days); CI runs = 50 only to check pipeline correctness, since the ~6–9 h LEAR/RF/XGB fit exceeds its timeout.

- Two constraints shaped the scope data-platform instability (source down for days, fetches slow and crash-prone, so we sometimes reduced features) and heavy compute neither of which affects the validity of the reported results.

- Each pillar has a caveat: the causal €3.54/MWh estimate assumes untestable unconfoundedness (plant availability, hydro levels missing); forecasts are DE-LU 2019–2024 only with year-averaged $rMAEs$ that tie LEAR and XGB ($p = 0.32$); and the K-Means regimes are descriptive (chosen k , interpreted labels), though the €62/MWh spread.



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